

Supporting Figures for Predicting growth, water-use efficiency and drought response through machine learning, GWAS and differential expression in Ponderosa pine

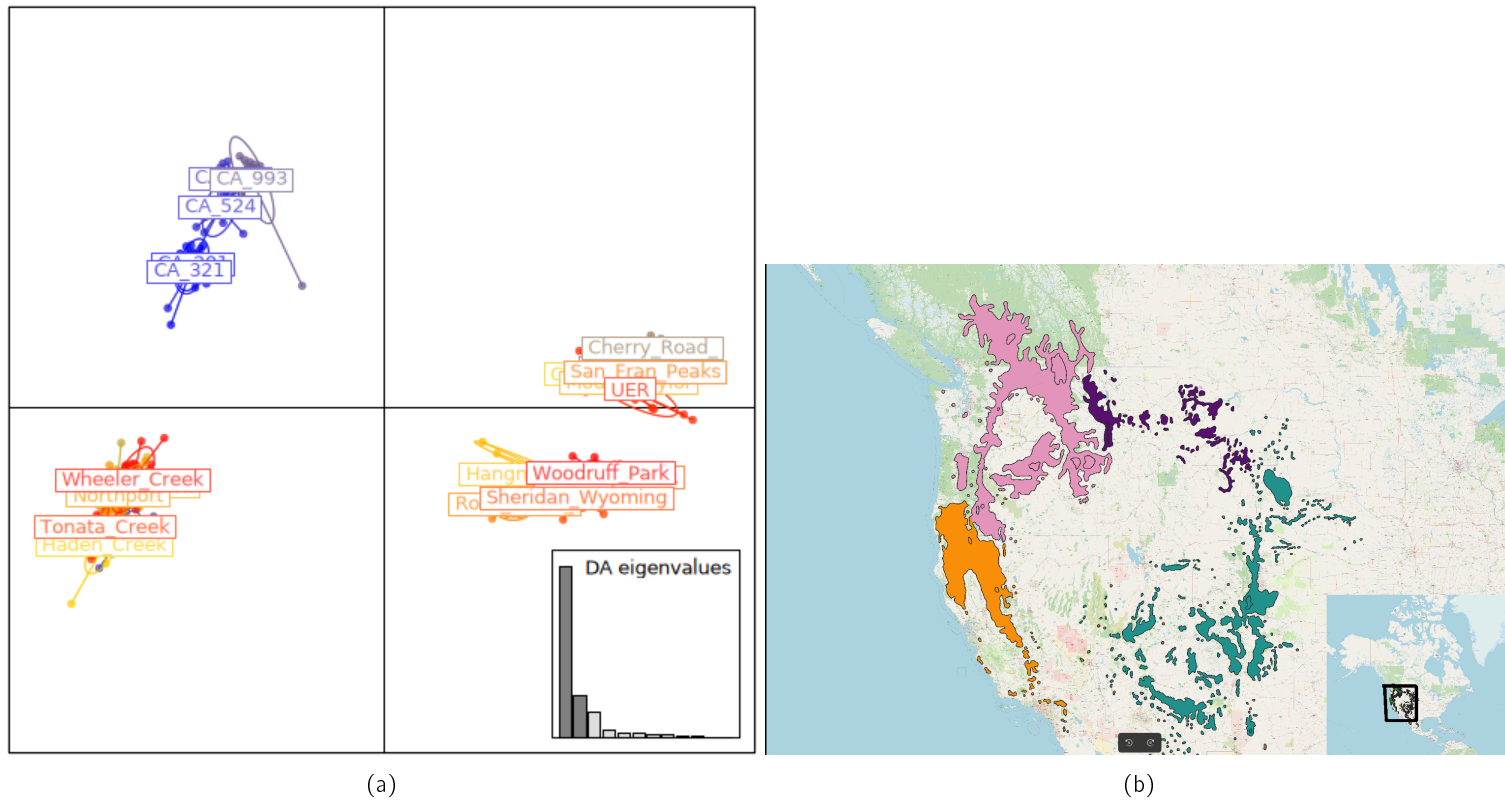


Figure S1. Population structure summary from Collins et al (2026). (A) DAPC of seed sources with an inset of variance explained across discriminant axes. (B) Map of ponderosa pine range with genetic clusters separated as colors seen in Figure 1.

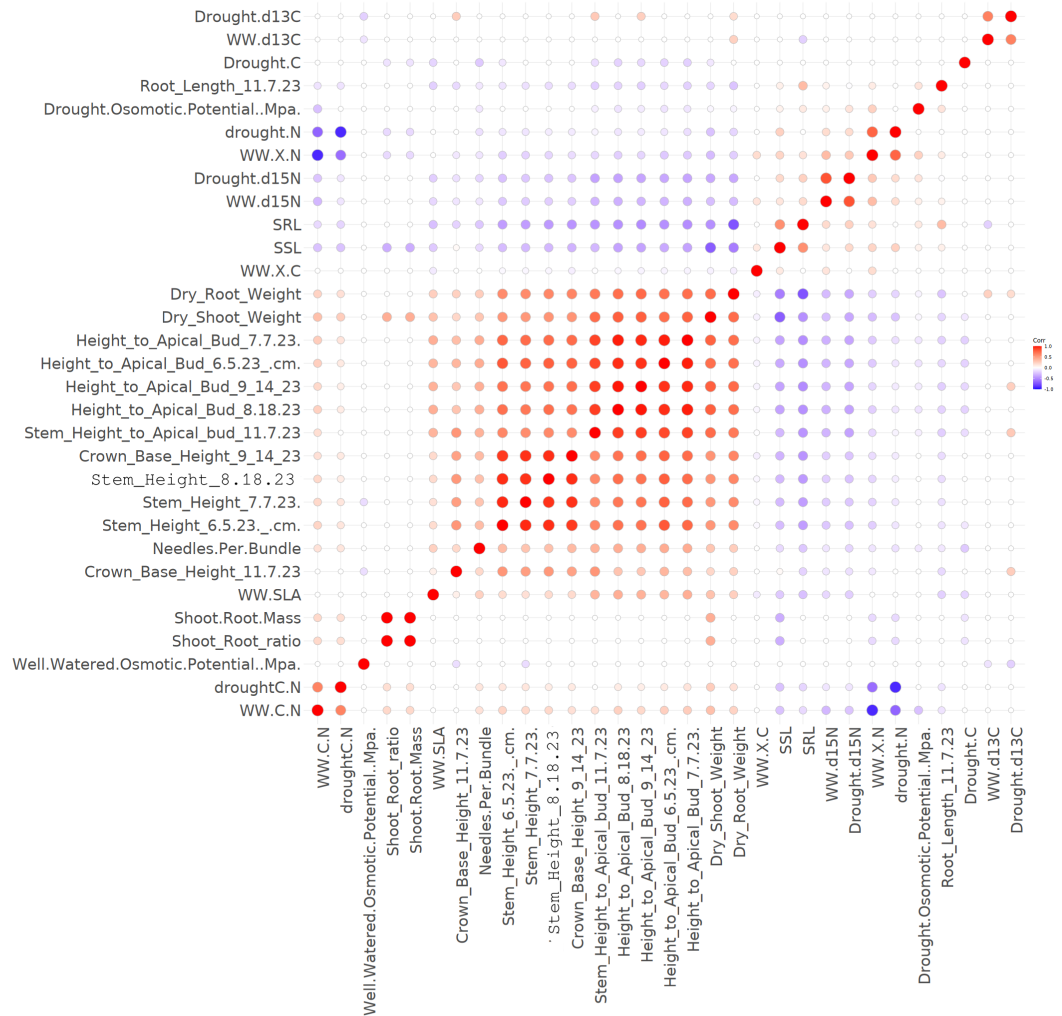


Figure S2.Trait Correlations. Spearman rank correlations of all traits. Non-significant correlations are shown via a blank white circle, while significant correlations are shown with either a blue or red circle indicating positive or negative, magnitude of correlation is shown by the intensity of color.

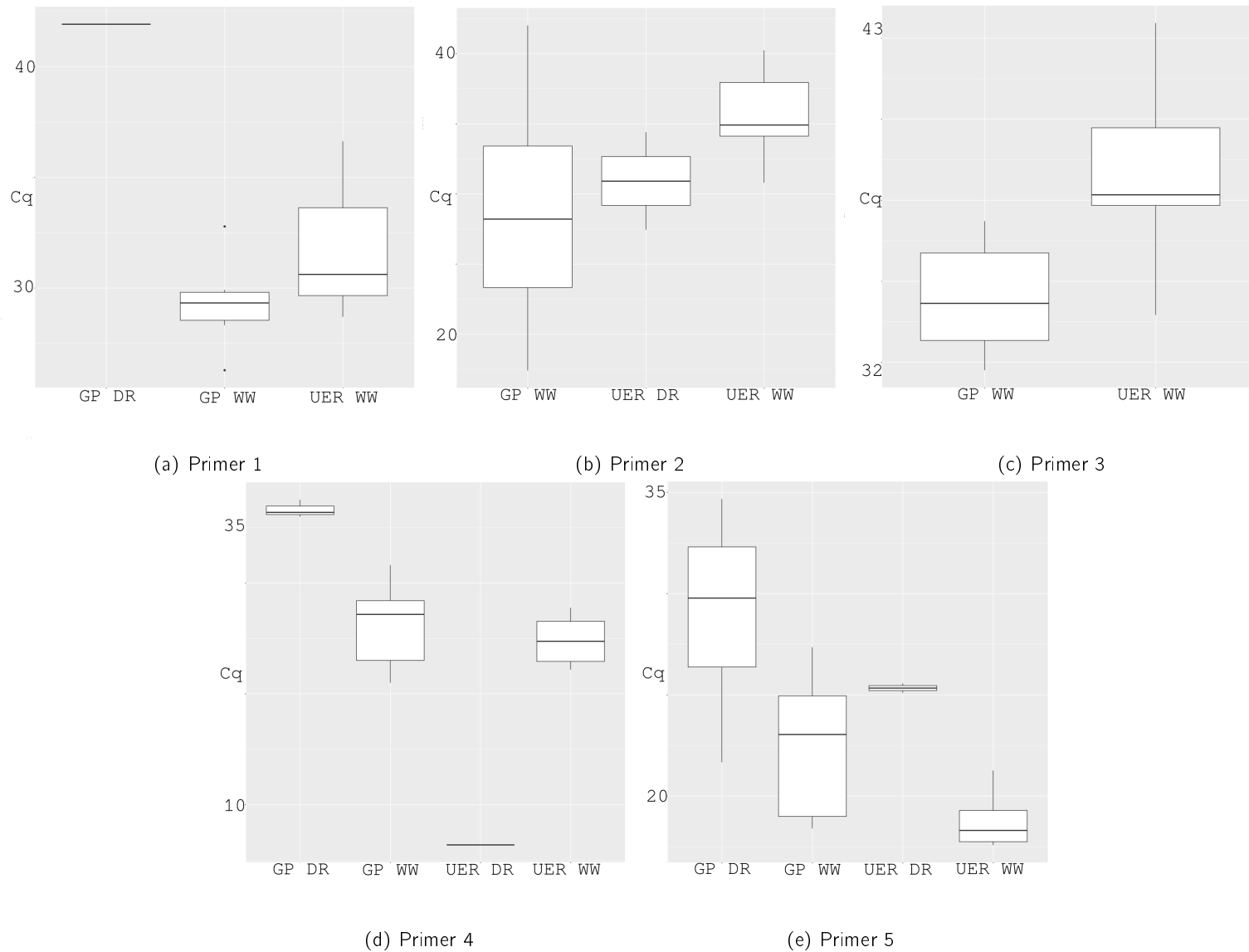


Figure S3.qPCR results for differential expression confirmation. Cq number for specific primers showing the expression level of transcripts chosen from the differential expression analysis. Seed sources (Green's Peak and Upper Enterprise Reservation) with respective treatment condition (well-watered and drought) are on the X axis. Seed sources are abbreviated Green's Peak is GP and Upper Enterprise Reservation is UER. Treatments are abbreviated as well, well-watered is WW and drought is DR.

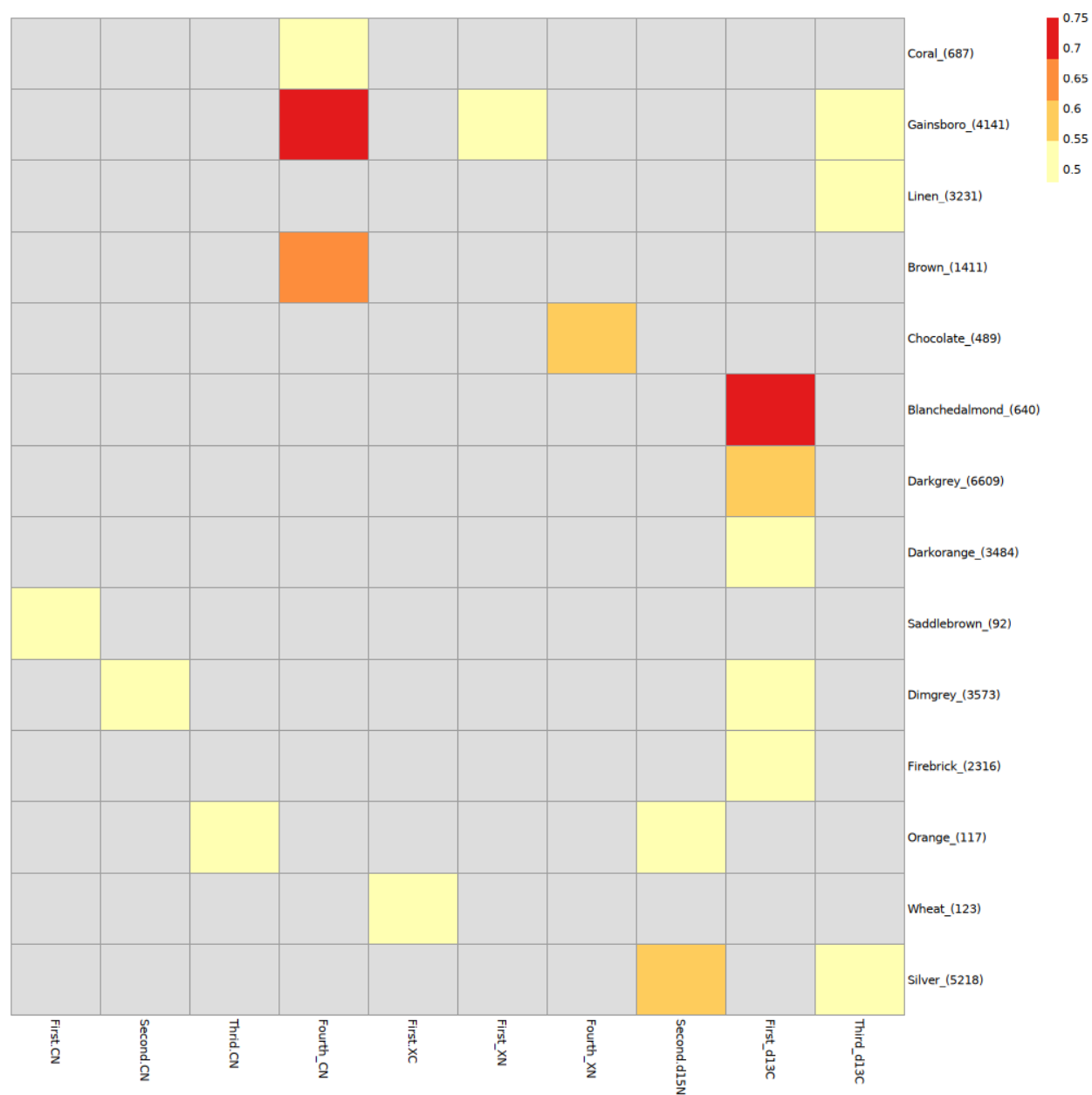


Figure S4. Heatmap of trait quartile and gene module correlations from WGCNA analysis. The key for correlation values is shown in the top right. Warmer colors indicate positive correlations and cooler colors indicate negative correlations. All non-gray boxes had a significant p-value.

